Names: Olivier Kwizera

Date: 16 October 2026

AmaliTech Apprenticeship Program

LAB2: TMDB Movie Data Analysis: Final Report

---

1. Introduction

This project builds a movie data analysis pipeline in Python. Movie data was fetched from The Movie Database (TMDb) API, cleaned and transformed with Pandas, and analysed through a set of key performance indicators (KPIs) covering financial performance, audience reception, and popularity. The analysis also compares franchise and standalone movies, identifies the most successful franchises and directors, and presents the main trends through visualizations built with Matplotlib.

2. Data Source

The data comes from the TMDb API (version 3), using the /movie/{id} endpoint. Each request included the append_to_response=credits parameter, which returns the cast and crew in the same response and avoids a second API call per movie.

The lab specified 19 movie IDs. One of them, ID 0, is not a valid TMDb movie and returned a 404 Not Found error. The pipeline handled this by logging the failed request and skipping that ID, so the final dataset contains 18 movies. The raw API responses were saved to data/raw/movies_raw.json so the analysis can be rerun without calling the API again.

The 18 movies are among the highest-grossing films ever released, including Avengers: Endgame, Avatar, Titanic, and Star Wars: The Force Awakens. They were released between 1997 and 2019.

3. Methodology

The project follows a modular structure. The logic lives in four Python modules (fetch.py, clean.py, kpis.py, visualize.py), and a Jupyter notebook (analysis.ipynb) runs the workflow step by step.

3.1 Data extraction.

A requests.Session was used to reuse the HTTP connection across requests. Authentication used the TMDb access token, with the API key as a fallback. Failed requests are caught and skipped so a single bad ID cannot stop the pipeline.

3.2 Data cleaning and preprocessing. The raw dataset had 18 rows and 28 columns. The following steps were applied:

- The irrelevant columns adult, imdb_id, original_title, video, and homepage were dropped.
- The JSON-like columns were flattened into readable text. The collection name was extracted from belongs_to_collection. Genres, production countries, production companies, and spoken languages were converted to pipe-separated strings (for example, “Adventure|Science Fiction|Action”). For spoken languages, the English name was used instead of the native name, so that “Español” appears as “Spanish”.
- Four new columns were derived from the credits field: cast (all cast names), cast_size, director, and crew_size. Movies with two directors, such as the Russo brothers, keep both names.
- The extracted columns were inspected with value_counts() to check for anomalies.
- budget, id, and popularity were converted to numeric types, with invalid values set to NaN. release_date was converted to datetime.
- Zero values in budget, revenue, and runtime were set to be replaced with NaN, since zero is not a realistic value for these fields. Budget and revenue were converted to millions of USD (budget_musd, revenue_musd).
- Movies with a vote count of 0 would have their average rating set to NaN, because a rating based on no votes is not meaningful.
- Placeholder text such as “No Data” in overview and tagline was set to be replaced with NaN.
- Duplicates were removed, rows with a missing ID or title were dropped, only rows with at least 10 non-null values were kept, and only movies with status “Released” were kept before the status column was dropped.
- Finally, the columns were reordered into the required layout and the index was reset.

The final cleaned dataset has 18 rows and 22 columns and was saved to data/processed/movies_clean.csv.

***3.3 KPI analysis**.*

Two derived metrics were added. Profit is revenue minus budget, and ROI (return on investment) is revenue divided by budget. A user-defined ranking function, rank_movies(), sorts the data by any metric, optionally applies a minimum threshold (such as budget of at least $10M for ROI, or at least 10 votes for ratings), and returns the top N results. All ten required rankings were produced with this single function. A reusable search_movies() function handles the filtered search queries, and grouped aggregations were used for the franchise, standalone, and director comparisons.

3.4 Visualization. Five charts were produced: Revenue vs. Budget, ROI Distribution by Genre, Popularity vs. Rating, Yearly Box Office Performance, and Franchise vs. Standalone comparison.

4. Data Quality Observations

The dataset turned out to be very clean. There were no zero values in budget, revenue, or runtime. No movie had a vote count of 0, and no placeholder text was found in the overview or tagline columns. All 18 movies had the status “Released”, so no rows were removed during filtering, and the dataset went from 18 rows to 18 rows. This is expected, because the selected movies are well-documented blockbusters.

A few things stood out during inspection:

- The API returned some columns not mentioned in the lab instructions (backdrop_path, origin_country, and softcore). They were excluded when the final column layout was applied.
- Some movies list less common spoken languages, such as Xhosa for Avengers: Endgame and Avengers: Infinity War. This reflects scenes set in the fictional nation of Wakanda, so it is not a data error.
- Only two of the 18 movies, Titanic and Beauty and the Beast, do not belong to a collection. This imbalance affects the franchise vs. standalone comparison (see Section 6).

5. Key Findings

5.1 Financial performance. Together, the 18 movies earned about $30,583 million in revenue on a combined budget of about $3,848 million. That is roughly 7.9 times their total budget.

KPI	Top movie	Value
Highest revenue	Avengers: Endgame	$2,927.45M
Highest budget	Avengers: Endgame	$356.00M
Highest profit	Avatar	$2,686.71M
Lowest profit	Star Wars: The Last Jedi	$1,034.41M
Highest ROI	Avatar	12.34x
Lowest ROI	Star Wars: The Last Jedi	4.45x

Avengers:

Endgame earned the most revenue, but it also had the largest budget. Avatar earned almost the same revenue ($2,923.71M) on a much smaller budget ($237M), which makes it both the most profitable movie and the one with the highest ROI. Titanic, released in 1997, ranks third in revenue and second in ROI (11.32x), which shows how well it has performed over time.

Every movie in the dataset was highly profitable. Even the lowest-profit movie, Star Wars: The Last Jedi, made over $1 billion in profit, and the lowest ROI was still 4.45x. The “worst” movies here are therefore only weak compared with other blockbusters, not in absolute terms.

***5.2 Audience reception and popularity**.*

The Avengers was the most voted movie (40,041 votes), followed by Avatar and Avengers: Infinity War. Avengers: Infinity War had the highest rating (8.24) and was also the most popular movie (popularity score 120.25). Four of the five most popular movies are Avengers films.

The lowest-rated movies were Jurassic World: Fallen Kingdom (6.52), Jurassic World (6.70), and Star Wars: The Last Jedi (6.73). Jurassic World is a good example of how revenue and rating can differ. It had the third-highest ROI (11.14x), but it is one of the lowest-rated films. Commercial success does not always mean critical approval.

***5.3 Search queries**.*

Search 1, for Science Fiction Action movies starring Bruce Willis, and Search 2, for movies starring Uma Thurman and directed by Quentin Tarantino, both returned no results. Neither actor appears in this set of 18 blockbusters, and no Tarantino films are included. The search functions work correctly but would need a larger dataset to return matches.

5.4 Franchise vs. standalone movies*.*

Metric	Franchise (16 movies)	Standalone (2 movies)
Mean revenue	$1,690.78M	$1,765.14M
Median ROI	7.84x	9.62x
Mean budget	$218.00M	$180.00M
Mean popularity	50.31	47.89
Mean rating	7.39	7.44

In this dataset, standalone movies had slightly higher mean revenue, a clearly higher median ROI, and a slightly higher rating, while spending less on average. Franchise movies had larger budgets and were slightly more popular. However, the standalone group contains only two movies, and one of them is Titanic, one of the most successful films ever made. These results should therefore not be taken to mean that standalone films generally outperform franchises.

5.5 Most successful franchises. The Avengers Collection was the most successful franchise by a wide margin. It had four movies, $7,904.08M in total revenue, and the highest mean rating of any franchise (7.95). The Star Wars Collection came second ($3,402.63M from two movies), followed by the Jurassic Park Collection ($2,982.01M). The Jurassic Park films were cheap to make relative to their earnings (mean budget $160M), but they had the lowest mean rating of any franchise (6.61). The Avatar Collection had the highest mean revenue per movie ($2,923.71M), but it has only one movie in this dataset.

5.6 Most successful directors. James Cameron directed the most commercially successful films, with $5,187.87M in total revenue from Avatar and Titanic. Anthony and Joe Russo followed with $4,979.86M from Avengers: Infinity War and Avengers: Endgame, and they had the highest mean rating of any director (8.22). Joss Whedon ($2,924.22M), and Chris Buck and Jennifer Lee ($2,727.90M from the two Frozen films) complete the top five. Co-directed films are credited to each director, so the Russo brothers and the Frozen directors appear with identical figures.

6. Visual Insights

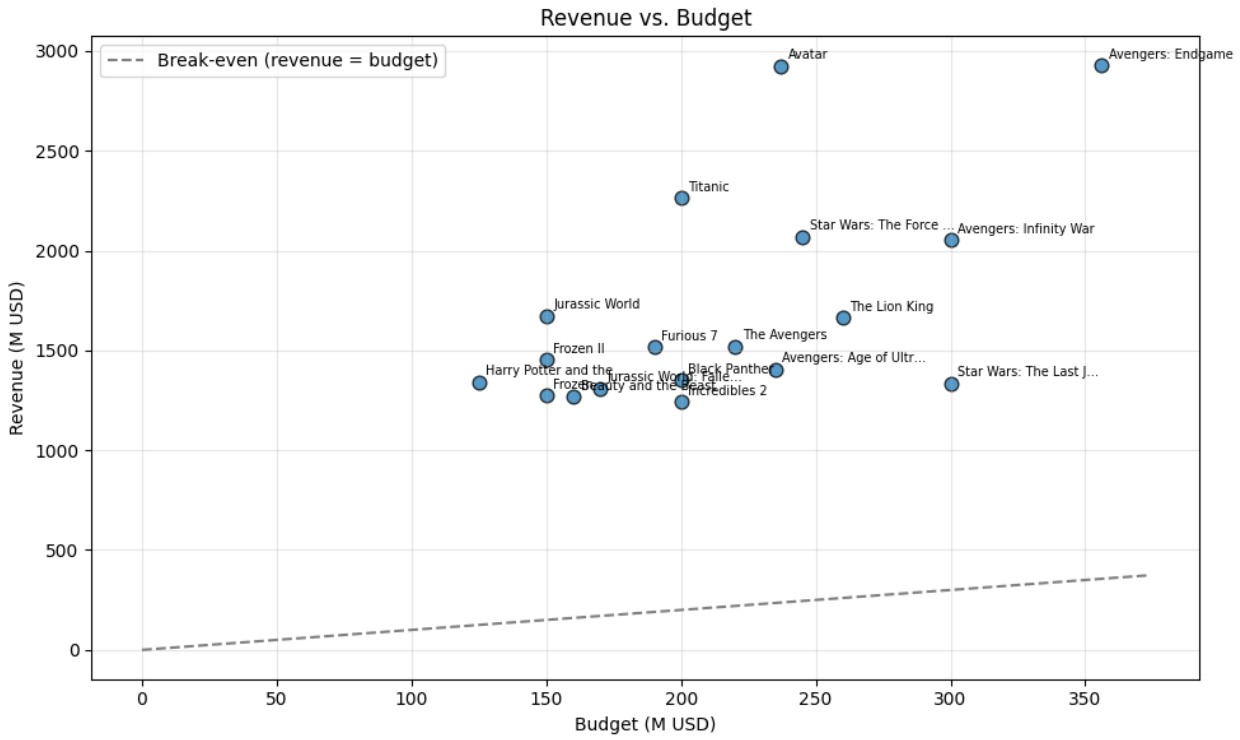


Figure 1: Revenue vs. Budget

Every movie sits far above the break-even line, so all 18 films earned many times their budget. There is no strong relationship between budget and revenue. Avengers: Endgame and Avatar earned almost the same amount despite a budget gap of more than $100M, and Star Wars: The Last Jedi had one of the largest budgets ($300M) but revenue in the lower half of the group.

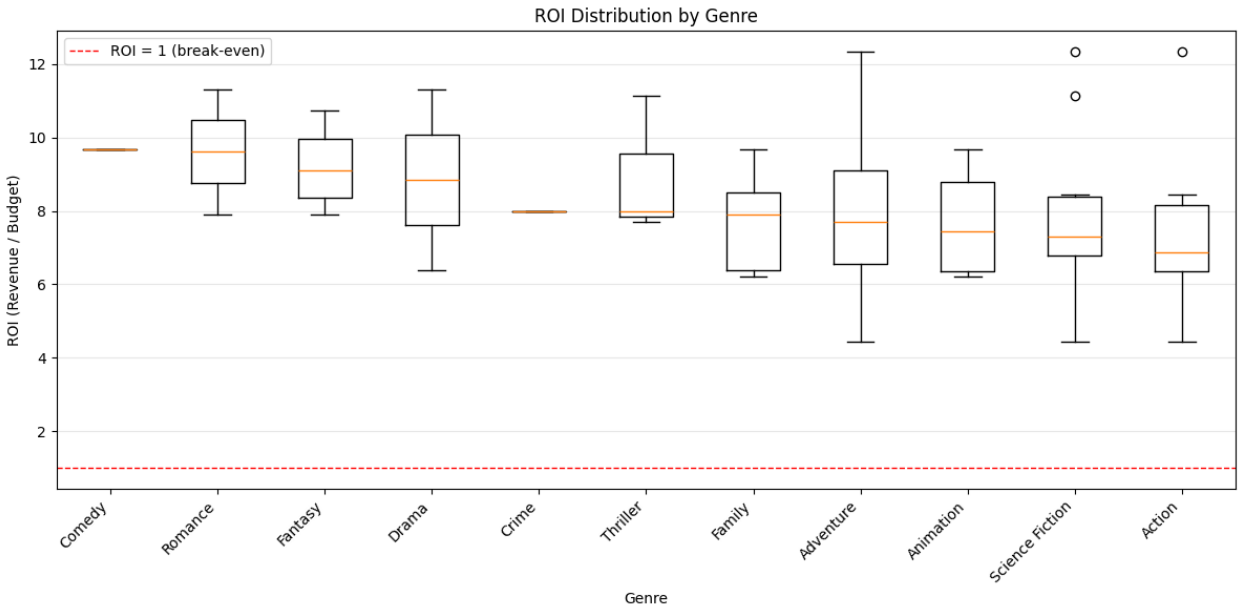


Figure 2: ROI Distribution by Genre

All genres have an ROI well above the break-even level of 1. Romance, Fantasy, and Comedy have the highest median ROI (roughly 9 to 10x), while Action and Science Fiction have the lowest (roughly 7x). Action and Science Fiction films tend to have bigger budgets, which lowers their ROI even when their revenue is high. Some genres, such as Comedy and Crime, are represented by a single movie, so their results are not reliable on their own.

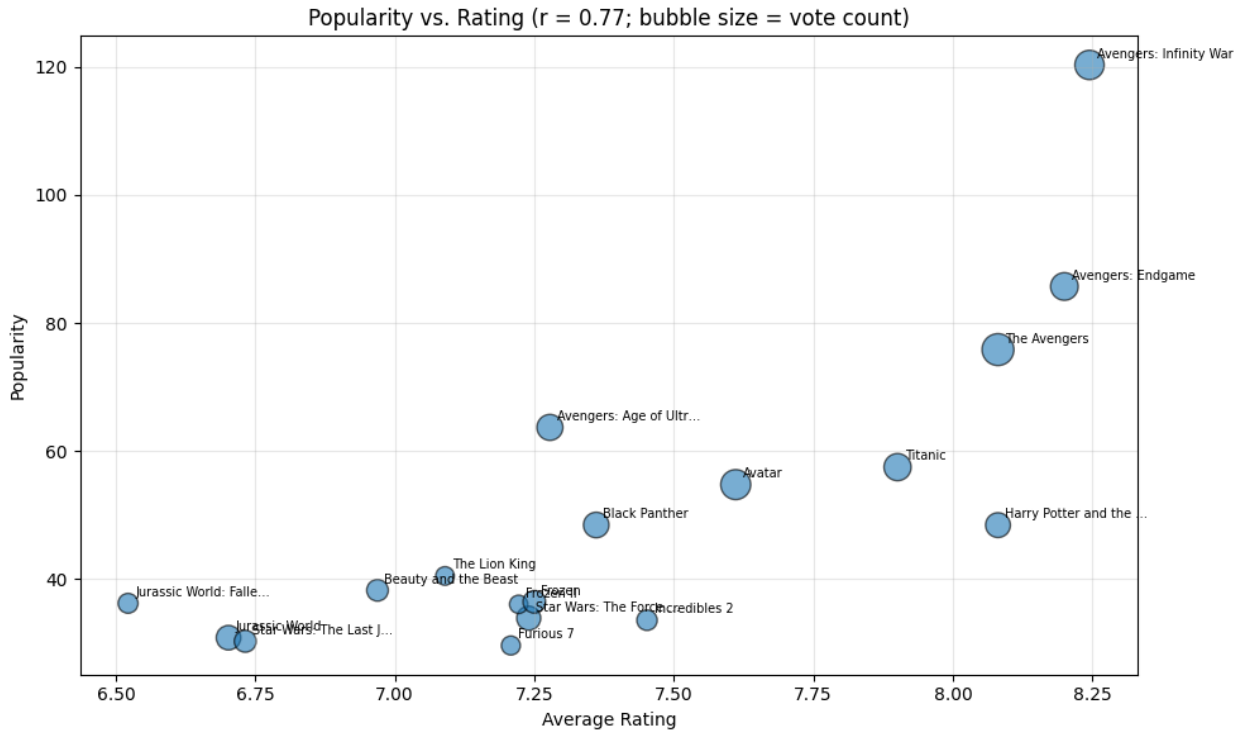


Figure 3: Popularity vs. Rating

Popularity and rating have a fairly strong positive correlation (r = 0.77), so higher-rated movies tend to be more popular. The Avengers films stand out in the top-right corner. TMDb popularity reflects current activity on the site, which partly explains why recent Marvel films score higher than older films with similar ratings, such as Titanic.

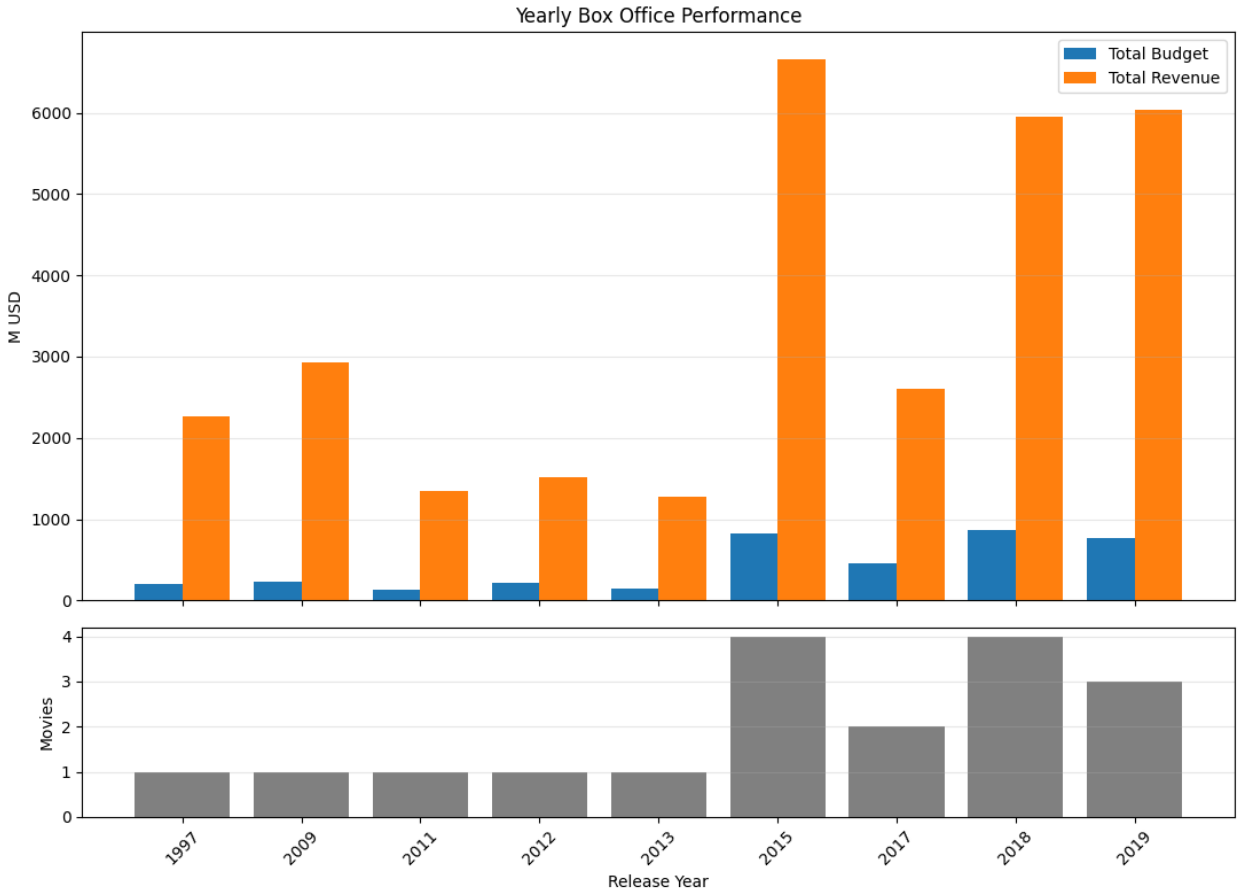


Figure 4: Yearly Box Office Performance

2015 had the highest total revenue (about $6,650M), followed by 2019 and 2018 (each about $6,000M). These years also have the most movies in the dataset (3 to 4 each), so the totals mainly reflect how many films from each year were selected, not a real change in the box office. In every year, revenue was far higher than budget.

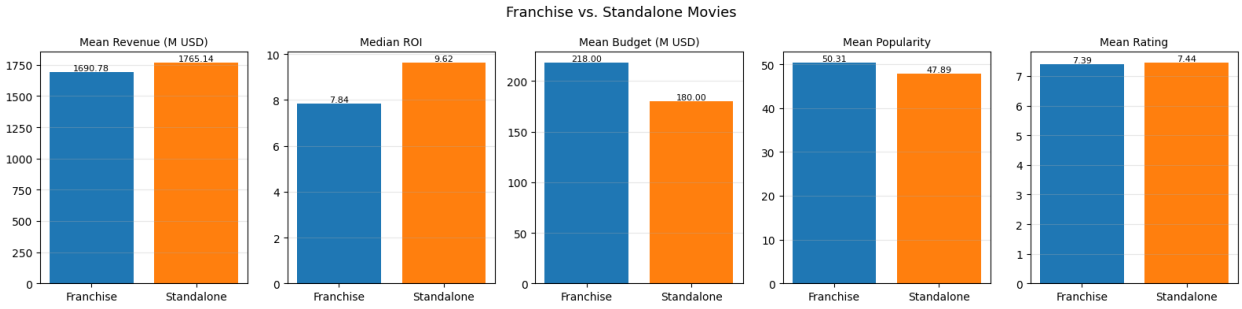


Figure 5: Franchise vs. Standalone Comparison

The chart shows the results from Section 5.4. Franchise movies have higher budgets and slightly higher popularity, while the two standalone movies lead in mean revenue, median ROI, and rating.

7. Limitations

- **Small sample.** With only 18 movies, the results describe this specific group of films and cannot be generalised to the film industry as a whole.
- **Selection bias.** The movies are among the highest-grossing ever made, so every film is profitable. The analysis cannot show what makes an average or failed movie different from a successful one.
- **Unbalanced groups.** There are 16 franchise movies and only 2 standalone movies, and several genres and franchises are represented by a single film. Comparisons involving these groups are fragile.
- **Unadjusted figures.** Revenue and budget are not adjusted for inflation, which disadvantages older films such as Titanic (1997).
- **Changing metrics.** Popularity, vote count, and rating change daily on TMDb, so rerunning the pipeline later may give slightly different numbers.
- **Empty searches.** Both search queries returned no results, because the requested actors and director are not in the dataset.

8. Conclusion

The pipeline fetched, cleaned, and analysed data for 18 blockbuster films. All of them were very profitable, earning about 7.9 times their combined budget. Avatar was the most efficient investment, with the highest profit and ROI, while Avengers: Endgame earned the most revenue. A bigger budget did not reliably lead to higher revenue or ROI. The Avengers Collection dominated as a franchise in both revenue and popularity, and James Cameron and the Russo brothers were the most commercially successful directors. Audience ratings were fairly closely linked to popularity, but not always to commercial success, as the Jurassic World films showed.

The modular design means the same pipeline could be applied to a much larger set of movies. A broader dataset, with inflation-adjusted figures and a balanced mix of franchise and standalone films, would allow stronger and more general conclusions.